

Relations and Functions

- **Definition 1**

Let X be a set. A **relation** $\sim$ on X is a rule that determines for every $x, y \in X$ whether the statement

$$x \sim y$$

is true or false. If it is true, we say that x is related to y . If it is false, we write $x \not\sim y$.

- **Example 1.1**

Let $X = \{1, 2, 3, 4\}$, the following are examples of relations on X .

1. The relation $\sim$ given by $1 \sim 2, 2 \sim 3, 3 \sim 1, 4 \sim 4$.
 2. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 2 \sim 4$.
 3. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4$.
-

- **Definition 2**

Let X be a set and $\sim$ a relation on X .

1. The relation is **reflexive** if $x \sim x$ for every $x \in X$.
2. The relation is **symmetric** if for every $x, y \in X$

$$x \sim y \implies y \sim x.$$

3. The relation is **antisymmetric** if for every $x, y \in X$

$$x \sim y \wedge y \sim x \implies x = y.$$

4. The relation is **transitive** if for every $x, y, z \in X$

$$x \sim y \wedge y \sim z \implies x \sim z.$$

- **Example 2.1**

On the set $\mathbb{Z}$ consider the relation

$$x \sim y \iff x \leq y.$$

Note that

1. $\sim$ is reflexive since $x \leq x$ for every $x \in \mathbb{Z}$.
 2. $\sim$ is not symmetric since $1 \leq 2$ but $2 \not\leq 1$.
 3. $\sim$ is antisymmetric because if $x \leq y$ and $y \leq x$ then $x = y$.
 4. $\sim$ is transitive since $x \leq y$ and $y \leq z$ implies that $x \leq z$.
-

- **Definition 3**

Let X be a set. An **equivalence relation** on X is a relation $\sim$ that satisfies the following properties.

1. $\sim$ is reflexive.
2. $\sim$ is symmetric.
3. $\sim$ is transitive.

- **Example 3.1**

Let $X = \{1, 2, 3, 4\}$. The following are examples of equivalence relations on X .

1. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4$.

2. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4, 2 \sim 3, 3 \sim 2, 1 \sim 4, 4 \sim 1$.
-

- **Definition 4**

Let X be a set. An **order relation** on X is a relation $\sim$ that satisfies the following properties.

1. $\sim$ is reflexive.
2. $\sim$ is antisymmetric.
3. $\sim$ is transitive.

- **Example 4.1**

Let $X = \{1, 2, 3, 4\}$. The following are examples of order relations on X .

1. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4$.
 2. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4, 1 \sim 2, 1 \sim 3, 1 \sim 4, 2 \sim 3, 2 \sim 4, 3 \sim 4$.
-

- **Definition 5**

A **function** $f: X \rightarrow Y$ is a correspondence of elements of X with elements of Y such that for all $x \in X$ there exists a unique $y \in Y$ such that $y = f(x)$.

$$\begin{aligned} f: X &\rightarrow Y \\ x &\mapsto y = f(x) \end{aligned}$$

The element $y = f(x)$ is called the **image** of x under f and the element x is called the **preimage** of y under f .

- **Definition 6**

If $f: X \rightarrow Y$ is a function, the **domain** of f denote $\text{Dom}(f)$ is X and the **codomain** of f is Y . The **range** of f , denoted $\text{Ran}(f)$, is the set

$$\text{Ran}(f) = \{y \in Y : y = f(x) \text{ for some } x \in X\}$$

Note that $\text{Ran}(f) \subseteq Y$.

- **Example 6.1**

1. The function $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$.
2. The function $g: \mathbb{R} \rightarrow \mathbb{R}, g(x) = \sin(x)$.
3. The function $h: [0, \infty) \rightarrow \mathbb{R}, h(x) = \sqrt{x}$.

- **Example 6.2**

Let $X = \{1, 2, 3\}$ and $Y = \{a, b, c\}$ the following examples of functions from X to Y .

1. The function $f: X \rightarrow Y$ defined by

$$f(1) = a \quad f(2) = b \quad f(3) = c.$$

2. The function $g: X \rightarrow Y$ defined by

$$g(1) = a \quad g(2) = a \quad g(3) = a.$$

3. The function $h: X \rightarrow Y$ defined by

$$h(1) = a \quad h(2) = b \quad h(3) = a.$$

- **Definition 7**

Two functions $f: X \rightarrow Y$ and $g: Z \rightarrow W$ are **equal** if $X = Z$, $Y = W$ and $f(x) = g(x)$ for all $x \in X$.

◦ **Example 7.1**

1. Let X be a set. The **identity function** on X is the function $\text{id}_x: X \rightarrow X$ given by $\text{id}_x(x) = x$.
2. Let X and Y be sets and $c \in Y$. The **constant function** with value c is the function $K_c: X \rightarrow Y$ given by $K_c(x) = c$.
3. Let X be a set and $A \subseteq X$. The **inclusion function** of A in X is the function $\iota: A \rightarrow X$ (or $\iota: A \hookrightarrow X$) given by $\forall x \in A, \iota(x) = x$.
4. If $f: X \rightarrow Y$ is a function and $A \subseteq X$, the **restriction** of f to A is the function $f|_A: A \rightarrow Y$ given by $f|_A(x) = f(x)$ for all $x \in A$.

• **Definition 8**

If $f: X \rightarrow Y$ and $g: Z \rightarrow W$ are functions, the **composition** of g with f , denoted $g \circ f$, is the function $g \circ f: X \rightarrow Z$ defined by $(g \circ f)(x) = g(f(x))$ for all $x \in X$.

◦ **Example 8.1**

Note that if $f: X \rightarrow Y$, f is a function and $A \subseteq X$, we have that $f|_A = f \circ \iota_A$ where $\iota: A \rightarrow X$.

$$A \xrightarrow{\iota} X \xrightarrow{f} Y.$$

• **Remark 8**

Note that even in the case both $g \circ f$ and $f \circ g$ are defined, the composition of two functions is not commutative, that is, in general $g \circ f \neq f \circ g$.

◦ **Example 8.2**

Consider the functions

$$\begin{aligned} f: \mathbb{R} &\rightarrow \mathbb{R}, & g: \mathbb{R} &\rightarrow \mathbb{R}. \\ x &\mapsto x^2, & x &\mapsto x + 1. \end{aligned}$$

Note that $g \circ f: \mathbb{R} \rightarrow \mathbb{R}$ and $f \circ g: \mathbb{R} \rightarrow \mathbb{R}$ with

$$\begin{aligned} g \circ f &= g(f(x)) & f \circ g &= f(g(x)) \\ &= g(x^2) & &= f(x + 1) \\ &= x^2 + 1 & &= (x + 1)^2. \end{aligned}$$

In particular if $x = 2$ we have that $(g \circ f)(2) = 5$ and $(f \circ g)(2) = 9$. Therefore $g \circ f \neq f \circ g$.

◦ **Proposition 8.3**

Let $f: X \rightarrow Y$, $g: Y \rightarrow Z$ and $h: Z \rightarrow W$ be functions, the $(h \circ g) \circ f = h \circ (g \circ f)$. In other words, the composition is associative.

■ **Proof 8.3.1**

Let $x \in X$ then

$$\begin{aligned} [(h \circ g) \circ f](x) &= (h \circ g)(f(x)) \\ &= h(g(f(x))) \\ &= h((g \circ f)(x)) \\ &= [h \circ (g \circ f)](x). \end{aligned}$$

Now, as $[(h \circ g) \circ f](x) = [h \circ (g \circ f)](x)$ for all $x \in X$ we have that $(h \circ g) \circ f = h \circ (g \circ f)$.

◦ **Example 8.4**

Let $s, t: \mathbb{Z} \rightarrow \mathbb{Z}$ be functions defined by $s(x) = x + 1$ and $t(x) = 2x$. Prove that $t \circ s \neq s \circ t$.

■ **Proof 8.4.1**

Let $x \in \mathbb{Z}$ then

$$\begin{aligned} t \circ s &= t(s(x)) & s \circ t &= s(t(x)) \\ &= t(x+1) & &= s(2x) \\ &= 2x+2 & &= 2x+1. \end{aligned}$$

In particular if $x = 2$ we have that $(t \circ s)(2) = 6$ and $(s \circ t)(2) = 5$. Therefore $t \circ s \neq s \circ t$.

○ **Example 8.5**

Let $X = \{1, 2, 3, 4, 5\}$ and let $f: X \rightarrow X$ be the function defined by

$$f(1) = 2, \quad f(2) = 2, \quad f(3) = 4, \quad f(4) = 4, \quad f(5) = 4.$$

Show that $f \circ f = f$. Find a function g such that $g \circ f = f$ and $f \circ g = f$.

■ **Solution 8.5.1**

$$\begin{aligned} &g: X \rightarrow X \\ g(1) &= 1, \quad g(2) = 2, \quad g(3) = 5, \quad g(4) = 4, \quad g(5) = 3. \end{aligned}$$

Then

$$\begin{aligned} (g \circ f)(1) &= g(f(1)) = g(2) = 2 = f(1), \\ (g \circ f)(2) &= g(f(2)) = g(2) = 2 = f(2), \\ (g \circ f)(3) &= g(f(3)) = g(4) = 4 = f(3), \\ (g \circ f)(4) &= g(f(4)) = g(4) = 4 = f(4), \\ (g \circ f)(5) &= g(f(5)) = g(4) = 4 = f(5). \end{aligned}$$

Therefore $g \circ f = f$. Similarly, we can prove $f \circ g = f$ and $f \circ f = f$.

• **Definition 9**

Let $f: X \rightarrow Y$ be a function.

We say that f is **injective** if given $x_1, x_2 \in X$, such that $f(x_1) = f(x_2)$ implies $x_1 = x_2$.

Equivalently, if given $x_1, x_2 \in X$ such that $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$.

We say that f is **surjective** if for every $y \in Y$ there exists $x \in X$ such that $y = f(x)$.

We say that f is **bijective** if it is both injective and surjective.

• **Remark 9**

If there exists a bijective function $f: X \rightarrow Y$. We say that X and Y are **one-to-one correspondence**.

○ **Example 9.1**

The identity function $\text{id}_x: X \rightarrow X$ is injective since if $\text{id}_x(x) = \text{id}_x(y)$ then $x = y$.

It is also surjective since for all $x \in X$ we have that $x = \text{id}_x(x)$.

Therefore, id_x is bijective.

○ **Example 9.2**

Consider the constant function $K_c: X \rightarrow Y$.

Note that if X has more than one element the function is not injective and if Y has more than one element the function is not surjective.

○ **Example 9.3**

The inclusion function $\iota: A \rightarrow X$ is injective since $\iota_A(x) = \iota_A(y)$ implies $x = y$. Note that, if $A = X$ then $\iota_x = \text{id}_x$. Therefore ι_A is surjective. If $A \neq X$ and is not surjective if $A \subset X$.

○ **Example 9.4**

Suppose that $f: A \rightarrow B$ and $g: B \rightarrow C$ are functions. Prove that if $g \circ f$ is injective then f is injective, and if $g \circ f$ is surjective then g is surjective.

■ **Solution 9.4.1**

Proof:

■ **First Part (If $g \circ f$ is injective, then f is injective):**

Let $x_1, x_2 \in A$ be arbitrary elements in the domain of f such that:

$$f(x_1) = f(x_2)$$

Since g is a function, applying g to both sides yields identical outputs:

$$g(f(x_1)) = g(f(x_2))$$

By definition of function composition, this is equivalent to:

$$(g \circ f)(x_1) = (g \circ f)(x_2)$$

We are given the hypothesis that $g \circ f$ is injective. By definition of an injective function, $(g \circ f)(x_1) = (g \circ f)(x_2) \implies x_1 = x_2$.

Thus, $f(x_1) = f(x_2) \implies x_1 = x_2$, proving that f is injective.

■ **Second Part (If $g \circ f$ is surjective, then g is surjective):**

Let $z \in C$ be an arbitrary element in the codomain of g .

We are given the hypothesis that $g \circ f: A \rightarrow C$ is surjective. By definition of a surjective function, there exists some element $x \in A$ such that:

$$(g \circ f)(x) = z$$

By definition of function composition, this means:

$$g(f(x)) = z$$

Let $y = f(x)$. Since $x \in A$ and $f: A \rightarrow B$, it follows that $y \in B$.

Substituting y back into the equation gives $g(y) = z$.

Thus, for any arbitrary $z \in C$, there exists an element $y \in B$ such that $g(y) = z$, proving that g is surjective.

○ **Example 9.5**

Prove that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x + 1$ is injective and surjective.

■ **Proof 9.5.1**

Injective Suppose that $f(x_1) = f(x_2)$, that is,

$$2x_1 + 1 = 2x_2 + 1$$

$$2x_1 = 2x_2$$

$$x_1 = x_2$$

then f is injective.

Surjective Let $y \in \mathbb{R}$ and consider $x = \frac{y-1}{2} \in \mathbb{R}$, then $f(x) = f\left(\frac{y-1}{2}\right)$
 $= 2\left(\frac{y-1}{2}\right) + 1 = y - 1 + 1 = y$.

Therefore f is surjective.

• **Definition 10**

Let $f: X \rightarrow Y$ be a function. If $A \subseteq X$ the **direct image** of A under f , denoted $f(A)$, is the set

$$f(A) = \{y \in Y : y = f(x) \text{ for some } x \in A\}$$

Note that $f(A) \subseteq Y$ and $f(X) = \text{Ran}(f)$.

If $B \subseteq Y$ the **inverse image** of B under f , denoted $f^{-1}(B)$, is the set

$$f^{-1}(B) = \{x \in X : f(x) \in B\}$$

◦ **Example 10.1**

Let $X = \{1, 2, 3, 4\}$, $Y = \{a, b, c, d\}$ and $f: X \rightarrow Y$ defined by $f(1) = a, f(2) = c, f(3) = c, f(4) = a$.

Consider $A = \{1, 2, 4\} \subseteq X$ and $B = \{a, b\} \subseteq Y$. Then

$$\begin{aligned} f(A) &= \{a, c\}, & f^{-1}(B) &= \{1, 4\} \\ f^{-1}(\{c\}) &= \{2, 3\}, & f^{-1}(\{d\}) &= \emptyset \end{aligned}$$

◦ **Proposition 10.2**

Let $f: X \rightarrow Y$ a function, $A, B \subseteq X$ and $C, D \subseteq Y$.

$$1. f(A \cap B) \subseteq f(A) \cap f(B).$$

Proof

Let $y \in f(A \cap B)$ then $y = f(x)$ for some $x \in A \cap B$. Since $x \in A \cap B$ then $x \in A$ and $x \in B$, that is, $y = f(x) \in f(A)$ and $y = f(x) \in f(B)$. Hence $y \in f(A) \cap f(B)$ and thus $f(A \cap B) \subseteq f(A) \cap f(B)$.

$$2. f(A \cup B) = f(A) \cup f(B).$$

Proof

($\subseteq$) Let $y \in f(A \cup B)$ then $y = f(x)$ for some $x \in A \cup B$. Since $x \in A \cup B$ then $x \in A$ or $x \in B$, that is, $y = f(x) \in f(A)$ or $y = f(x) \in f(B)$ hence $y \in f(A) \cup f(B)$ and thus $f(A \cup B) \subseteq f(A) \cup f(B)$.

($\supseteq$) Let $y \in f(A) \cup f(B)$ then $y \in f(A)$ or $y \in f(B)$, that is, $y = f(a)$ for some $a \in A$ or $y = f(b)$ for some $b \in B$. Therefore $y \in f(A \cup B)$ since $a, b \in A \cup B$ and thus $f(A) \cup f(B) \subseteq f(A \cup B)$.

$$3. f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D).$$

Proof

($\subseteq$) Let $x \in f^{-1}(C \cap D)$ then $f(x) \in C \cap D$, that is, $f(x) \in C$ and $f(x) \in D$ hence $x \in f^{-1}(C)$ and $x \in f^{-1}(D)$ therefore $x \in f^{-1}(C) \cap f^{-1}(D)$ and thus $f^{-1}(C \cap D) \subseteq f^{-1}(C) \cap f^{-1}(D)$.

($\supseteq$) Let $x \in f^{-1}(C) \cap f^{-1}(D)$ then $x \in f^{-1}(C)$ and $x \in f^{-1}(D)$, that is, $f(x) \in C$ and $f(x) \in D$ hence $f(x) \in C \cap D$, that is, $x \in f^{-1}(C \cap D)$ and thus $f^{-1}(C) \cap f^{-1}(D) \subseteq f^{-1}(C \cap D)$.

$$4. f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D).$$

Proof

($\subseteq$) Let $x \in f^{-1}(C \cup D)$ then $f(x) \in C \cup D$, that is, $f(x) \in C$ or $f(x) \in D$ hence $x \in f^{-1}(C)$ or $x \in f^{-1}(D)$ therefore $x \in f^{-1}(C) \cup f^{-1}(D)$ and thus $f^{-1}(C \cup D) \subseteq f^{-1}(C) \cup f^{-1}(D)$.

($\supseteq$) Let $x \in f^{-1}(C) \cup f^{-1}(D)$ then $x \in f^{-1}(C)$ or $x \in f^{-1}(D)$, that is, $f(x) \in C$ or $f(x) \in D$ hence $f(x) \in C \cup D$, that is, $x \in f^{-1}(C \cup D)$ and thus $f^{-1}(C) \cup f^{-1}(D) \subseteq f^{-1}(C \cup D)$.

◦ **Remark 10.2**

Note that $f(A) \cap f(B) \subseteq f(A \cap B)$ is not always true, for example consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$, $A = \{-1\}$ and $B = \{1\}$ then $f(A) = 1 = f(B)$. But $f(A \cap B) = f(\emptyset) = \emptyset$.

◦ **Exercise 10.3**

If $f: X \rightarrow Y$ is injective and $A, B \subseteq X$ then $f(A) \cap f(B) \subseteq f(A \cap B)$.

◦ **Theorem 10.4**

Let $f: X \rightarrow Y$ be an injective function. Let $y \in \text{Ran}$ then there exists a **unique** $x \in X$ such that $y = f(x)$, that is, $f^{-1}(\{y\}) = \{x\}$.

■ **Proof 10.4.1**

Let $y \in \text{Ran}(f)$ then there exists $x \in X$ such that $y = f(x)$. Suppose that there exists $x' \in X$ such that $y = f(x')$ then $f(x) = f(x')$ and as f is injective we have that $x = x'$.

• **Definition 11**

A function $f: X \rightarrow Y$ is called **invertible** if there exists a function $g: Y \rightarrow X$ such that $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$. Such a function is called **inverse** of f .

• **Remark 11**

If $f: X \rightarrow Y$ is invertible then its inverse is unique. In fact, suppose that $g, g': Y \rightarrow X$ are both inverse of f , then

$$g \circ f = \text{id}_X = g' \circ f \text{ and } f \circ g = \text{id}_Y = f \circ g'$$

hence $g = \text{id}_X \circ g = (g' \circ f) \circ g = g' \circ (f \circ g) = g' \circ \text{id}_Y = g'$.

If f is invertible, the inverse of f is denoted f^{-1} . Note that for all $x \in X$ and $y \in Y$

$$y = f(x), \text{ if and only if, } f^{-1}(y) = x$$

◦ **Example 11.1**

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be the functions.

$$f(x) = \begin{cases} x^2 + 2 & (x \leq 0) \\ 2 - x^2 & (x > 0) \end{cases} \quad g(x) = \begin{cases} -\sqrt{x-2} & (x \geq 2) \\ \sqrt{2-x} & (x < 2) \end{cases}$$

Prove that $g = f^{-1}$.

■ **Proof 11.1.1**

We need to prove that $g \circ f = \text{id}_{\mathbb{R}}$, that is, $(g \circ f)(x) = x$ for all $x \in \mathbb{R}$.

If $x \leq 0$ then $x^2 \geq 0$ and $x^2 + 2 \geq 2$, hence $(g \circ f)(x) = g(f(x)) = g(x^2 + 2) = -\sqrt{(x^2 + 2) - 2} = -\sqrt{x^2} = -|x| = -(-x) = x$.

If $x > 0$ then $x^2 > 0$ and $2 - x^2 < 2$, hence $(g \circ f)(x) = g(f(x)) = g(2 - x^2) = \sqrt{2 - (2 - x^2)} = \sqrt{x^2} = |x| = x$.

◦ **Theorem 11.2**

A function $f: X \rightarrow Y$ is invertible, if and only if, f is bijective.

■ **Proof 11.2.1**

($\Rightarrow$) Suppose that f is invertible then there exists $f^{-1}: Y \rightarrow X$ such that $f^{-1} \circ f = \text{id}_X$ and $f \circ f^{-1} = \text{id}_Y$.

If $f(x) = f(x')$ then

$$\begin{aligned} f^{-1}(f(x)) &= f^{-1}(f(x')) \\ (f^{-1} \circ f)(x) &= (f^{-1} \circ f)(x') \\ \text{id}_X(x) &= \text{id}_X(x') \\ x &= x' \end{aligned}$$

hence f is injective.

Let $y \in Y$ then $y = \text{id}_Y(y) = (f \circ f^{-1})(y) = f(f^{-1}(y))$ with $f^{-1}(y) \in X$, hence f is surjective. Therefore, f is bijective.

($\Leftarrow$) TBD

- **Definition 12**

Let $f: A \rightarrow B$ and $g: B \rightarrow A$ be two functions.

g is a **left inverse** of f if $g \circ f = \text{id}_A$.

g is a **right inverse** of f if $f \circ g = \text{id}_B$.